

EUNGWANG SEO

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EDUCATION

University of Glasgow, United Kingdom Ph.D. in Physics and Astronomy <i>Thesis: Gravitational Lensing of Gravitational Waves: From Searches to Cosmology and Population Inference</i> <i>Advisor: Martin Anthony Hendry and Graham Woan</i>	2022-2026 (expected)
The Chinese University of Hong Kong, Hong Kong M.Phil. in Physics <i>Thesis: Detecting and Interpreting Lensed Gravitational Waves Using Bayesian Inference</i> <i>Advisor: Tjonnje Guang Feng Li and Otto Akseli Hannuksela</i>	2019-2022
Kyunghee University, South Korea B.S. in Astronomy and Space Science, <i>summa cum laude (valedictorian)</i>	2013-2019

AWARDS AND HONORS

- Graduate School Mobility Funding Scholarship, University of Glasgow (£3k)	2024
- Peter MacGuire Bequest (now Lord Kelvin Fund), University of Glasgow (£2k)	2023
- College of Science and Engineering Scholarship (Full funding: tuition + stipend), University of Glasgow	2022
- President's Award (Valedictorian), Kyunghee University	2019
- KSSS Award for honored graduates, The Korean Space Science Society	2019
- Dean's list, Kyunghee University	2013-2018

PRESENTATIONS

<Invited Presentations>

1. KASI 76th Workshop on GW and NR , Daejeon, Republic of Korea Search for gravitational wave lensing signatures in the first part of LIGO-Virgo-KAGRA's fourth observing run	May 2026
2. KU Leuven GW Seminar , Leuven, Belgium Strong lensing of dark sirens and galaxies as a probe of the Hubble constant	Oct. 2025
3. UC Louvain GW Seminar , Louvain-la-Neuve, Belgium Direct measurement of the Hubble constant from strongly lensed dark sirens via the Hubble–Lemaître law	Oct. 2025
4. Korea Astronomy and Space Science Institute Seminar , Daejeon, Republic of Korea Detecting and interpreting lensed gravitational waves using Bayesian inference	Apr. 2024
5. Chinese University of Hong Kong Summer School , Hong Kong SAR Gravitational-wave strong lensing and microlensing	Aug. 2023
6. Korean Gravitational Wave Group monthly Colloquium , Seoul, Republic of Korea Searching for microlensing signatures in gravitational waves	Nov. 2021

<Contributed Presentations>

1. GWTC-4.0 CBC Lensing Webinar , Online GWTC-4.0: Searches for Gravitational-Wave Lensing Signatures	Apr. 2026
2. 13th Belgium-Dutch Gravitational-Wave Meeting , Nijmegen, the Netherlands Residual test to search for microlensing signatures in strongly lensed gravitational waves	Oct. 2025
3. National Astronomy Meeting 2025 , Durham, England Detection of microlensing imprints in strongly lensed gravitational wave events using residual analysis	Jul. 2025
4. SUPA Cormark Meeting 2023 , Glasgow, Scotland Rejection sampling for mapping model-independent parameters to lens-model-dependent properties	Dec. 2023
5. Amaldi15 , Online Inferring properties of dark galactic halos using strongly lensed gravitational waves	Jul. 2023

<Poster Presentations>

1. **GR24 / Amaldi16, Glasgow, Scotland** Jul. 2025
Lens-model-independent analysis of strongly lensed and microlensed gravitational waves
2. **LIGO-Virgo-KAGRA Collaboration Meeting, Melbourne, Australia** Mar. 2025
Residual test to search for microlensing signatures in strongly lensed gravitational waves
3. **Theo Murphy meeting: Multi-messenger Gravitational Lensing, Manchester, England** Mar. 2024
Constraining lens model parameters of strongly lensed gravitational waves with multiple lens models
4. **LIGO-Virgo-KAGRA Collaboration Meeting, Toyama, Japan** Sep. 2023
Gravitational lensing aided luminosity distance estimation for compact binary coalescences
5. **National Astronomy Meeting 2023, Cardiff, Wales** Jul. 2023
Reducing distance uncertainties for binary black holes by accounting for lensing
6. **LIGO-Virgo-KAGRA Collaboration Meeting, Chicago, IL, USA** Mar. 2023
Inferring properties of dark galactic halos using strongly lensed gravitational waves
7. **Asia-Pacific School and Workshop on Gravitation and Cosmology 2020, Daejeon, South Korea** Feb. 2020
Estimating dark halo mass using lensed gravitational waves

WORK EXPERIENCE AND SERVICE

- Analysis & Review roles** (*Analyst; Editorial team; P&P, code, and result reviewer*) 2021-present
- LIGO-Virgo-KAGRA Collaboration and Lensing Group
- Paper Reviewer** 2020-present
- Physical Review D
- Visiting Researcher** (*worked with Prof. Jose Maria Ezquiaga*) 2024
- Niels Bohr Institute, Copenhagen
- Research Assistant** (*worked with Prof. Chunglee Kim and Dr. Kyungmin Kim*) 2022
- Ewha Womans University, Seoul
- Undergraduate Internship** (*supervised by Prof. Hyung Mok Lee*) 2017
- Seoul National University, Seoul

TEACHING EXPERIENCE

- Graduate Teaching Assistant**
1. School of Physics and Astronomy, University of Glasgow Jan. 2023 - Jan. 2026
- *Astronomy 1 Lab. and Astronomy 2 Lab.*
 2. Department of Physics, The Chinese University of Hong Kong Sep. 2019 - Jun. 2021
- *Electromagnetic Theory I, Introduction to Astronomy and Astrophysics, and Statistical Mechanics*

EXTRACURRICULAR ACTIVITIES

- Student President of the Department of Astronomy and Space Science, Kyunghee University** 2017
- Full-time Military Service, Republic of Korea Navy** 2014 - 2016

TECHNICAL COMPETENCIES

- Programming:** C++, Fortran90, IDL, Python
- Software & Tools:** **GW Data Analysis:** BILBY, GWPopulation, GWpy, PyCBC
GW Lensing: LINK (golum + gravelmaps)
Cosmology: gwcsmo, COLOSSUS, Pyhalo